

Imperial School of Banking & Management Studies

Session Plan – PGDM 2025–27 | Semester III

Accredited | Approved by AICTE | Affiliated to Savitribai Phule Pune University

Course Title	Agentic AI & Advanced Analytics in Finance
Course Code	PGDM-SEM3-SPEC-AIFINANCE
Specialization	Finance (Elective / Specialization Subject)
Programme	PGDM 2025–27 Batch — Semester III
Faculty	Tejas Jadhav CFA Charterholder, FRM Certified
Total Teaching Hours	30 Hours (20 Sessions × 1.5 Hours)
Session Plan Submitted On	June 2026
Academic Year	2025–26 (Semester III)

About the Course

Agentic AI & Advanced Analytics in Finance is a 30-hour specialization course for Semester III PGDM students at ISBMS, designed to bridge theoretical finance and contemporary AI-driven industry practice. The course equips students with production-grade skills in demand at Global Capability Centres (GCCs) and financial institutions across India and globally. Students explore agentic AI architectures, prompt engineering, Retrieval-Augmented Generation (RAG) pipelines, algorithmic trading, credit and market risk modelling (PD models, Monte Carlo VaR), regulatory compliance automation (Basel III/IV), and fund administration workflows. Drawing from Harvard Case Method, NUS sandbox-first learning, GARP's FRM framework, and CFA Institute standards, each module culminates in a deployable project artifact that builds a live portfolio. The capstone simulates a real GCC hiring panel, ensuring students are placement-ready upon completion.

Word count: ~130 words

Course Objectives (Bloom's Taxonomy)

- To **recall** key concepts of Large Language Models, AI agent architectures, and financial data APIs as applied in financial services operations. **L1 – Remember**

2. To **explain** the principles of Retrieval-Augmented Generation (RAG), Model Context Protocol, and multi-agent orchestration within the context of financial workflow automation.
L2 – Understand
3. To **apply** prompt engineering, Python programming, and quantitative modelling techniques in hands-on lab sessions to solve real-world financial problems. **L3 – Apply**
4. To **analyse** risk metrics (VaR, PD), regulatory compliance requirements (Basel III/IV), and algorithmic trading performance using AI-assisted analytical tools. **L4 – Analyse**
5. To **evaluate** enterprise AI governance frameworks, model risk management policies, and the regulatory and ethical implications of deploying AI in financial institutions.
L5 – Evaluate
6. To **design** an integrated financial AI system—encompassing a fund administration module, risk analytics engine, and compliance report generator—demonstrating placement-ready skills for GCC and financial sector roles. **L6 – Create**

Session-Wise Teaching Plan (30 Hours)

Sess.	Hrs	Topic / Sub-Topic	Learning Objectives	Pedagogy & Activity	Teaching Aids & References
Module 1: Foundations of Agentic AI in Financial Services 3 Hours					
1	1.5 h	Introduction to AI in Finance: LLM fundamentals, transformer architecture, financial NLP use-cases; evolution from rule-based to agentic systems	Recall LLM concepts; explain financial NLP applications (L1, L2)	Lecture + Live Demo; Quiz (financial NLP examples)	Jadhav, <i>Claude AI for Finance Professionals</i> (Ch. 1); GARP FRM Study Notes – Technology Risk
2	1.5 h	Agentic AI Architecture: Agent loops (observe–plan–act), financial data APIs (Bloomberg, NSE), tool calling; Lab: Build a structured financial data extraction prompt	Apply prompt engineering to extract structured financial data (L3)	Sandbox Lab (Python + OpenAI/Claude API); Case: "How Morgan Stanley uses GPT-4 for wealth advisors"	LangChain Docs; Anthropic API Reference; <i>Artificial Intelligence in Finance</i> , Yves Hilpisch (Ch. 2)
Module 2: Prompt Engineering & RAG for Financial Analysis 3 Hours					
3	1.5 h	Prompt Engineering for Finance: Zero-shot, few-shot, chain-	Apply prompt patterns; evaluate quality of AI	Lecture + Live Prompting; Group Activity:	Jadhav, <i>Claude AI for Finance Professionals</i> (Ch.

Sess.	Hrs	Topic / Sub-Topic	Learning Objectives	Pedagogy & Activity	Teaching Aids & References
		of-thought prompting; financial reasoning patterns; common failure modes and hallucination risks	financial outputs (L3, L5)	redesign a failing financial prompt	3–4); White, J. et al. (2023). "A Prompt Pattern Catalog" – ArXiv:2302.11382
4	1.5 h	RAG Pipeline Construction: Vector databases (ChromaDB/ Pinecone), document chunking, embeddings, retrieval; Lab: Build a 200-page prospectus Q&A bot	Design and deploy a RAG pipeline for unstructured financial documents (L6)	Hands-On Lab; Industry Case: "SEBI DRHP Analysis using RAG"	LlamaIndex Documentation; Lewis et al. (2020). "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks" – NeurIPS 2020
Module 3: Model Context Protocol & Agent Orchestration 3 Hours					
5	1.5 h	Model Context Protocol (MCP): Architecture, tool definitions, memory management, context windows; financial context design patterns	Explain MCP architecture; apply context design in financial scenarios (L2, L3)	Lecture + Architecture Diagram Workshop; Simulation: design a credit analyst agent	Anthropic MCP Specification (2024); Jadhav, <i>Claude AI for Finance Professionals</i> (Ch.6)
6	1.5 h	Multi-Agent Orchestration: CrewAI / LangGraph frameworks; agent roles (researcher, analyst, writer); Lab: Build a 3-agent investment memo pipeline	Design multi-agent pipelines; evaluate output quality of orchestrated agents (L6, L5)	Sandbox Lab; Case: "Multi-agent due diligence at a PE firm"	CrewAI Documentation; Wu et al. (2023). "AutoGen: Enabling Next-Gen LLM Applications via Multi-Agent Conversation" – ArXiv:2308.08155
Module 4: Algorithmic Trading & Execution Strategies 3 Hours					
7	1.5 h	Algorithmic Trading Fundamentals: Market microstructure, order types, trading signals (momentum, mean-reversion), backtesting	Explain algorithmic trading concepts; analyse signal performance metrics (L2, L4)	Lecture + Case Discussion: "Knight Capital \$440M algo failure"; Simulation exercise	Chan, E. (2013). <i>Algorithmic Trading: Winning Strategies and Their Rationale</i> (Chs. 1–3); SEBI Circular on Algorithmic Trading (2012)

Sess.	Hrs	Topic / Sub-Topic	Learning Objectives	Pedagogy & Activity	Teaching Aids & References
		methodology; regulatory considerations (SEBI algo trading guidelines)			
8	1.5 h	AI-Enhanced Execution Strategies: Reinforcement learning in trading, NLP-based signal generation from news; Lab: Build an AI trading bot with backtested results (Backtrader + Python)	Apply AI to trading strategy design; evaluate backtested Sharpe ratio and drawdown (L3, L4)	Hands-On Lab (Python/ Backtrader); Peer Review of backtesting results	Hilpisch, Y. (2020). <i>Artificial Intelligence in Finance</i> (Ch. 7); QuantLib Python Documentation
Module 5: Risk Modelling – PD Models & Monte Carlo VaR 3 Hours					
9	1.5 h	Credit Risk & Probability of Default Models: Logistic regression, ML-based PD estimation, scorecard development; validation metrics (Gini, KS statistic, AUC-ROC)	Apply ML models to credit risk; analyse model performance metrics (L3, L4)	Lecture + Python Demo (scikit-learn); Case: "Fintech lender PD model failure"	Basel Committee on Banking Supervision. (2006). <i>Basel II: International Convergence</i> ; Siddiqi, N. (2017). <i>Intelligent Credit Scoring</i> (Ch. 4)
10	1.5 h	Market Risk & Monte Carlo VaR: Historical simulation vs Monte Carlo, 10,000-path simulation, Conditional VaR (CVaR), risk decomposition; Lab: Generate a VaR report with AI commentary	Design Monte Carlo VaR model; evaluate risk outputs against Basel thresholds (L6, L5) — Assignment topic delivered in this session	Sandbox Lab (NumPy/ Python); Industry Simulation: multi-asset portfolio VaR	GARP FRM Part II – Market Risk Measurement; Hull, J. C. (2022). <i>Options, Futures, and Other Derivatives</i> (Ch. 21–22)
Module 6: RegTech & Automated Basel III/IV Compliance 3 Hours					
11	1.5 h	RegTech Overview & Basel III/IV: Regulatory capital	Recall Basel III/IV framework; explain AI's role in	Lecture + Regulatory Simulation	BIS Basel Committee. (2017). <i>Basel III: Finalising</i>

Sess.	Hrs	Topic / Sub-Topic	Learning Objectives	Pedagogy & Activity	Teaching Aids & References
		adequacy (CET1, Tier 1, Total Capital ratios), LCR, NSFR, leverage ratio; AI applications in compliance monitoring	regulatory compliance (L1, L2)	(AxiomSL-style templates); Guest Talk segment (video)	<i>Post-Crisis Reforms</i> ; Deloitte RegTech Universe Report (2023)
12	1.5 h	Automated Compliance Reporting: Document generation pipelines, structured data extraction from financial statements; Lab: Auto-generate a Basel III Pillar 3 disclosure document using Claude API	Design automated compliance reporting pipeline; evaluate output accuracy (L6, L5)	Hands-On Lab (Claude API + Python); Peer Review of generated disclosure	Anthropic Claude API Documentation; EY RegTech Report: "AI in Regulatory Reporting" (2023)
Module 7: Enterprise AI Governance, Security & Cost Management 3 Hours					
13	1.5 h	Enterprise AI Governance: Model Risk Management (SR 11-7), AI audit frameworks, explainability requirements (SHAP/LIME), bias and fairness in financial AI; EU AI Act implications	Evaluate AI governance frameworks; analyse regulatory risk in AI deployment (L5, L4)	Case Discussion: "Goldman Sachs Apple Card gender bias investigation"; Governance Blueprint Workshop	Federal Reserve SR 11-7 Guidance; FSB. (2022). "Artificial Intelligence and Machine Learning in Financial Services"
14	1.5 h	AI Security & Cost Management: Prompt injection attacks, hallucination risk mitigation, Total Cost of Ownership (TCO) modelling for LLM deployment; Lab: Design an AI governance blueprint document — Case Study conducted in this session	Design AI governance blueprint; evaluate security and cost trade-offs (L6, L5)	Red-Team Exercise (prompt injection); TCO Modelling Spreadsheet Lab	OWASP LLM Top 10 (2023); Jadhav, <i>Claude AI for Finance Professionals</i> (Ch.8)

Sess.	Hrs	Topic / Sub-Topic	Learning Objectives	Pedagogy & Activity	Teaching Aids & References
Module 8: Fund Administration – NAV & Transfer Agency Automation 3 Hours					
15	1.5 h	Fund Accounting & NAV Automation: Fund administration lifecycle (trade capture → NAV → investor reporting), AI in valuation, corporate actions processing; industry landscape (Opus, HC Global, SS&C)	Explain fund administration workflows; apply AI to NAV calculation process (L2, L3)	Lecture + Industry Case: "Opus Fund Services AI transformation"; Role-play exercise	IOSCO Principles for Fund Valuation (2013); Citi Securities Services: "AI in Fund Administration" Report (2023)
16	1.5 h	Transfer Agency & KYC Automation: Investor onboarding (AML/KYC), transaction processing, e-KYC using NLP; Lab: Generate a daily NAV report with AI-written commentary	Design an automated NAV reporting system; evaluate KYC automation accuracy (L6, L4)	Hands-On Lab (Python + Claude API); Peer Review of NAV report	KFINTECH Product Documentation; SEBI Circular on KYC Registration Agencies (2022)
Module 9: Market Commentary & Valuation Dashboards 3 Hours					
17	1.5 h	Market Commentary Generation: NLP for financial news analysis, sentiment scoring (FinBERT), earnings call transcript analysis, AI-generated research reports	Apply NLP models to market analysis; analyse sentiment signals from financial text (L3, L4)	Live Demo (FinBERT); Case: "AI-generated earnings summaries at Bloomberg Intelligence"	Yang et al. (2020). "FinBERT: A Pre-trained Financial Language Representation Model for Financial Text Mining" – IJCAI 2020; Bloomberg NLP Research
18	1.5 h	Valuation Dashboards & Data Visualization: Streamlit/Python dashboards, DCF modelling with AI assistance, interactive charts (Plotly), market intelligence	Design and deploy an interactive financial dashboard; evaluate usability and data accuracy (L6, L5)	Hands-On Lab (Streamlit + Plotly); Live Demo	Streamlit Documentation; Damodaran, A. (2012). <i>Investment Valuation</i> (Ch. 2); Plotly Python Documentation

Sess.	Hrs	Topic / Sub-Topic	Learning Objectives	Pedagogy & Activity	Teaching Aids & References
		prototype; Lab: Build a Streamlit market dashboard			
Module 10: GCC Capstone – Integrated Financial AI System 3 Hours					
19	1.5 h	Capstone Preparation: Portfolio integration (combine modules 1–9 outputs), pitch deck design (problem → solution → architecture → demo → impact), peer review and rehearsal	Design integrated AI system; evaluate project readiness against GCC hiring criteria (L6, L5)	Workshop (peer review panels); Instructor feedback rounds	McKinsey GCC India Report (2024); Jadhav, <i>Claude AI for Finance Professionals</i> (Capstone Chapter)
20	1.5 h	GCC Capstone Presentation: Mock hiring panel (simulated GCC interviewers), live demo of deployed project, Q&A, course debrief, placement readiness assessment	Present integrated system to mock panel; demonstrate placement-ready communication (L6)	Mock Hiring Panel; Peer Evaluation; Course Debrief Discussion	NASSCOM GCC Trends Report (2024); Course Portfolio Rubric

Assessments

Assignment: AI-Powered Monte Carlo VaR Risk Report

Introduced in: Session 10 | **Submitted to Academic Coordinator:** At least 2 sessions (3 hours) before due date

Conducted: In-class (Session 10 lab component) + take-home writeup

Description: Students will develop a Python-based Monte Carlo Value-at-Risk (VaR) model for a hypothetical equity portfolio of five NSE-listed stocks. The assignment requires students to: (1) collect 2 years of historical price data using Yahoo Finance API, (2) run 10,000 Monte Carlo simulation paths, (3) calculate 95% and 99% VaR and CVaR, (4) generate an AI-written risk commentary using Claude API, and (5) interpret results with reference to Basel III market risk capital requirements.

Deliverables:

- Jupyter Notebook with all code, charts, and simulation output
- 2-page PDF summary report with AI-generated commentary and regulatory interpretation

Criterion	Model Accuracy & Code	AI Integration Quality	Regulatory Interpretation	Report Presentation	Total
Marks	30	30	20	20	100

Case Study: "EisnerAmper GCC Pune – Automating Audit Analytics with Generative AI"

Conducted: Session 14 (in-class, 45 minutes discussion + 30 minutes write-up)

Submitted to Academic Coordinator: Case material distributed and submitted to coordinator at least 2 sessions (3 hours) before the session

Case Summary: EisnerAmper's Global Capability Centre in Pune adopted generative AI tools to automate document review, extract financial data from unstructured PDFs, and draft preliminary audit findings. The case explores: technology stack selection, change management, regulatory risk (PCAOB/ICAI audit standards), ROI measurement, and competitive implications for mid-tier audit firms.

Discussion Questions:

- What specific audit tasks were automated, and which AI techniques were applied to each?
- Analyse the governance and data privacy risks EisnerAmper faced in deploying AI in audit workflows.
- Using the AI governance framework from Module 7, evaluate EisnerAmper's approach. What controls were missing?
- How does this GCC model create competitive moats, and what does it mean for entry-level finance careers?

Written Submission: Individual 1-page analytical response (submitted within 48 hours)

Criterion	Case Comprehension	Analytical Depth	Framework Application	Written Communication	Total
Marks	25	35	25	15	100

References

Textbooks (Primary)

1. Hilpisch, Y. (2020). *Artificial Intelligence in Finance: A Python-Based Guide*. O'Reilly Media, Sebastopol, CA. ISBN: 978-1492055433. **Textbook**
2. Chan, E. P. (2013). *Algorithmic Trading: Winning Strategies and Their Rationale*. Wiley Trading, Hoboken, NJ. ISBN: 978-1118460146. **Textbook**
3. Hull, J. C. (2022). *Options, Futures, and Other Derivatives* (11th ed.). Pearson Education, Upper Saddle River, NJ. ISBN: 978-0136939917. **Textbook**
4. Jadhav, T. (2024). *Claude AI for Finance Professionals: Agentic Workflows, RAG, and Automation*. Amazon KDP. [Amazon Best Seller – AI & Finance Category] **Textbook**

Research Papers (Primary)

1. Yang, Z., Dai, Z., Yang, Y., Carbonell, J., Salakhutdinov, R., & Le, Q. V. (2019). *XLNet: Generalized Autoregressive Pretraining for Language Understanding*. Advances in Neural Information Processing Systems (NeurIPS 2019). ArXiv:1906.08237. **Research Paper**
2. Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., ... Kiela, D. (2020). *Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks*. NeurIPS 2020. ArXiv:2005.11401. **Research Paper**
3. White, J., Fu, Q., Hays, S., Sandborn, M., Olea, C., Gilbert, H., ... Schmidt, D. C. (2023). *A Prompt Pattern Catalog to Enhance Prompt Engineering with ChatGPT*. ArXiv:2302.11382. **Research Paper**
4. Financial Stability Board (FSB). (2022). *Artificial Intelligence and Machine Learning in Financial Services: Market Developments and Financial Stability Implications*. FSB Report, November 2022. **Research Paper**

Institutional & Regulatory References

1. Basel Committee on Banking Supervision. (2017). *Basel III: Finalising Post-Crisis Reforms*. Bank for International Settlements (BIS), December 2017.
2. GARP. (2024). *FRM Exam Part II Study Guide: Market Risk Measurement and Management*. Global Association of Risk Professionals.
3. NASSCOM. (2024). *GCC in India: Talent, Technology and Transformation*. NASSCOM Research Report.
4. Federal Reserve System. (2011). *SR 11-7: Guidance on Model Risk Management*. Board of Governors of the Federal Reserve System.

90 Relevant Topics – Specialization Subject List

As required under Guideline 7: Submitted by 13th August 2026. Topics span all 10 modules and adjacent industry domains.

1. Transformer architecture and attention mechanism	2. Pre-training and fine-tuning of LLMs	3. Tokenization and embedding models	Category A: Large Language Models & AI Foundation (Topics 1–9)
6. Open-source vs. proprietary LLMs in finance	7. Multimodal AI in financial document analysis	8. AI-assisted financial research and report generation	
			4. LLM hallucination and mitigation strategies
			9. FinBERT domain-specific language models
10. Zero-shot prompting in financial contexts	11. Few-shot and chain-of-thought prompting	12. Structured output prompting (JSON / tables)	Category B: Prompt Engineering & RAG (Topics 10–18)
15. Document chunking and embedding strategies	16. RAG pipeline evaluation metrics	17. DRHP and prospectus analysis using RAG	
			13. System prompt design for financial agents
			18. Agent architectures: RAG vs. standard F
19. Agent loop: observe–plan–act–reflect	20. Model Context Protocol (MCP) specification	21. Tool calling and function execution in agents	Category C: Agentic AI &

			Orchestration (Topics 19–27)
24. Investment memo generation pipeline (3-agent)	25. Autonomous financial research agents	26. Human-in-the-loop agent design	22. Multi-agent framework: CrewAI and LangGraph
			27. Agent evaluation: quality assurance
28. Market microstructure and order book dynamics	29. Momentum and mean-reversion trading strategies	30. Pairs trading and statistical arbitrage	Category D: Algorithmic Trading (Topics 28–36)
33. Reinforcement learning for trading (DQN, PPO)	34. NLP-based trading signal generation	35. High-frequency trading (HFT) and latency	31. Backtesting: methodology and overfitting risks
			36. SEBI algorithmic trading regulations
37. Value-at-Risk (VaR): historical and parametric	38. Monte Carlo simulation for market risk	39. Conditional Value-at-Risk (CVaR / Expected Shortfall)	Category E: Risk Analytics & Modelling (Topics 37–45)
42. Credit scorecard development with ML	43. Stress testing and scenario analysis	44. Model validation: Gini,	40. Probability of Default

		KS statistic, AUC-ROC	model development
			45. AI-generated commentaries and reports
46. Basel III capital adequacy framework (CET1, Tier 1)	47. Liquidity Coverage Ratio (LCR) and NSFR	48. Pillar 3 disclosure automation	Category F: RegTech & Compliance Automation (Topics 46–54)
51. FATF recommendations and RegTech compliance	52. Automated SAR (Suspicious Activity Report) filing	53. AxiomSL and Wolters Kluwer OneSumX platforms	
			49. Anti-Money Laundering (AML) and detection
			54. RBI guidelines in banking
55. Model Risk Management (SR 11-7 framework)	56. Explainable AI (XAI): SHAP and LIME methods	57. Algorithmic bias and fairness in credit decisions	Category G: AI Governance & Ethics (Topics 55–63)
60. Data privacy in AI: GDPR and DPDP Act (India)	61. AI Total Cost of Ownership (TCO) modelling	62. Responsible AI principles for financial institutions	
			58. EU AI Act and financial services implications

			63. AI audit model documentation standards
64. Net Asset Value (NAV) calculation lifecycle	65. Corporate actions processing and automation	66. Transfer Agency operations and e-KYC	Category H: Fund Administration & FinOps (Topics 64–72)
69. Reconciliation automation in fund operations	70. Alternative investment fund (AIF) administration	71. SEBI mutual fund regulations and compliance	67. Fund accounting standards 9, ASC 820
			72. AI in portfolio performance attribution
73. Sentiment analysis with FinBERT	74. Earnings call transcript mining	75. News-based event-driven trading signals	Category I: Market Analytics & Visualization (Topics 73–81)
78. ESG data integration in valuation models	79. Streamlit dashboard development for finance	80. Plotly and interactive financial charting	76. Discounted Cash Flow (DCF) model with AI
			81. Real-time market data APIs (NSE, BSE, FRED)

82. Global Capability Centres (GCC): India landscape 2024	83. GCC hiring criteria in finance and AI roles	84. Portfolio design for finance + AI job applications	Category J: GCC Operations Capstone & Career Readiness (Topics 82– 90)
87. Technical interview preparation: Python + Finance	88. Capstone: Integrated Financial AI System design	89. Industry mentorship and alumni network leverage	85. Pitch d design for technical finance pro
			90. Emerg trends: Quantum computing finance

Faculty: Tejas Jadhav (CFA, FRM)	HOD / Programme Coordinator	Academic Coordinator: Mr. Bhushan Bhegade	Principal / Director
----------------------------------	-----------------------------	---	----------------------